

Saivenkat Gattu

Data Engineer

Location: St louis, MO | Phone: +1 314-661-0091 | Email: saivenkatgattu7@gmail.com | [LinkedIN](#)

Summary

- Over 4+ years of experience as a Data Engineer, designing and implementing robust data pipelines and scalable data architectures.
- Skilled in programming languages including Python, Java, Scala, R, and SQL, with hands-on expertise in big data technologies such as Apache Hadoop, Spark, and Kafka.
- Proficient in ETL development and data integration using tools like Talend and Informatica to process and transform data from diverse sources.
- Extensive experience in data warehousing with platforms like Snowflake, Redshift, and BigQuery, enhancing query performance and storage efficiency.
- Adept in deploying containerized applications with Docker and Kubernetes, and automating workflows using CI/CD tools like Jenkins and GitHub Actions.
- Strong understanding of cloud platforms (AWS, Azure, GCP) and data governance frameworks (HIPAA, GDPR), ensuring compliance and security.
- Experienced in creating real-time analytics dashboards using Power BI, Grafana, and Kafka, enabling data-driven decision-making.
- Highly skilled in data modeling, predictive analytics, and implementing innovative solutions to optimize operational efficiency and support business goals.

Skills

- **Programming Languages:** Python, Java, Scala, R, SQL
- **Big Data Technologies:** Apache Hadoop, Apache Spark, Kafka
- **Database Systems:** MySQL, PostgreSQL, SQL Server, MongoDB
- **Data Integration Tools:** Talend, Informatica
- **Cloud Platforms:** AWS, Google Cloud Platform, Microsoft Azure
- **Data Warehousing:** Snowflake, Redshift, BigQuery
- **DevOps Tools:** Docker, Kubernetes, Jenkins, GitHub Actions
- **Infrastructure as Code:** Terraform, Ansible
- **Data Modeling:** OLAP, OLTP, Data Lakes
- **Version Control:** Git, Bitbucket
- **Scripting:** Bash, Shell scripting
- **Project Management Tools:** Jira, Trello, Asana
- **Data Governance Frameworks:** GDPR, HIPAA

Work Experience

PNC Bank

Aug 2024 -Current

Data Engineer

- Designed and deployed real-time data processing pipelines using Apache Kafka and Spark to enhance operational analytics and reporting.
- Developed comprehensive Redshift-based data warehousing solutions to streamline analytical workflows and reduce query execution times.
- Built robust ETL workflows using Python to extract, transform, and load data from diverse sources, ensuring data accuracy and completeness.
- Implemented HIPAA-compliant data security measures across the data ecosystem, safeguarding sensitive information and maintaining compliance.
- Conducted detailed root cause analysis for data inconsistencies, developed preventive strategies, and implemented corrective actions to ensure data reliability.

Dell Technologies

Jan 2019 – Jun 2022

Data Engineer

- Designed and implemented scalable and efficient data lakes on Microsoft Azure to support big data storage and analytics.
- Built and optimized Apache Spark workflows to process large datasets, ensuring timely availability of analytics.
- Developed and maintained ETL pipelines using Talend and Informatica to integrate and transform data from multiple sources.
- Migrated enterprise data warehouses to Snowflake, improving overall query performance and reducing data latency.
- Deployed containerized data services using Docker and orchestrated workflows with Kubernetes to improve scalability and reliability.

- Automated CI/CD pipelines for data engineering projects using Jenkins and GitHub Actions to streamline deployment processes.
- Designed and implemented data models for OLAP systems to enhance reporting and support better decision-making.
- Created monitoring dashboards using Grafana and Power BI to identify and resolve pipeline issues proactively.
- Collaborated with business stakeholders and technical teams to gather requirements and deliver actionable insights.

Trigent

Jan 2018 – Feb 2019

Data Engineer

- Designed and implemented robust ETL pipelines using Talend to automate data migration and transformation processes, reducing manual intervention.
- Analysed large-scale datasets utilizing Apache Hadoop and Spark to generate actionable insights for business-critical operations.
- Developed Python-based data cleaning and validation scripts, improving overall data accuracy and readiness for analytics.
- Maintained and optimized PostgreSQL databases, ensuring seamless performance and minimal downtime for data access.
- Integrated Apache Kafka to enable real-time data streaming and improve system responsiveness across multiple applications.
- Automated routine operational tasks using advanced Bash scripting, significantly reducing manual efforts and increasing efficiency.
- Collaborated with cross-functional teams to establish data governance processes and ensure compliance with GDPR and other regulations.

Projects

Data Migration and Integration Framework

- Designed an advanced data integration framework using Talend and Informatica to migrate legacy systems to Snowflake, achieving seamless data transitions and minimal downtime.
- Developed Python-based data validation scripts and automated processes to ensure zero data loss during migration, reducing errors and improving reliability.

Real-Time Analytics Dashboard

- Built comprehensive real-time analytics dashboards leveraging Apache Kafka and Power BI to provide actionable insights and operational transparency for stakeholders.
- Deployed backend services on AWS infrastructure using Kubernetes clusters, ensuring high availability and scalability of the analytics platform.

Predictive Maintenance System

- Developed machine learning models using R and Python to predict equipment failures, reducing downtime and increasing operational efficiency by leveraging advanced analytics techniques.
- Integrated PostgreSQL and BigQuery for efficient storage and retrieval of predictive analytics data, ensuring scalability and ease of access for large datasets.

Education

Master of Science (M.S.) in Information Technology

Saint Louis University, St. Louis, Missouri

Bachelor of Engineering (B.E.) in Computer Science and Engineering

Koneru Lakshmaiah University, Guntur, Andhra Pradesh, India